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3. The table shows how the solubility of potassium nitrate in water varies with temperature.

Temperature (°C)	Solubility (g per 100 g of water)
0	13
10	21
20	32
30	46
40	64
50	86

(a) Tick (✓) the box that shows approximately how many times more soluble potassium nitrate is at 40 °C than it is at 10 °C. [1]

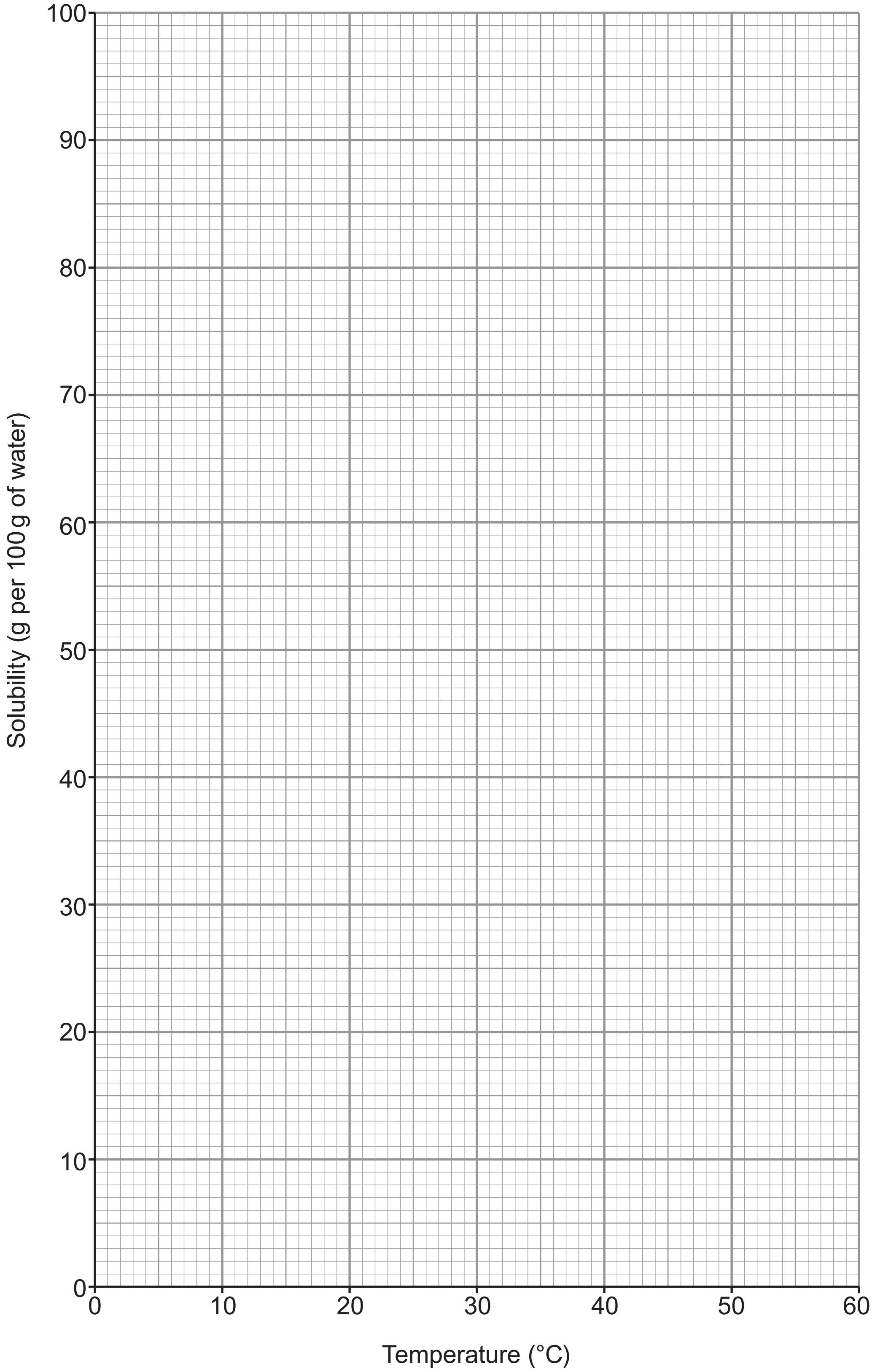
2 times ☐

3 times ☐

4 times ☐

5 times ☐

(b) (i) Plot the solubility of potassium nitrate against temperature on the grid.
Draw a suitable line. [3]



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(ii) I. State the solubility of potassium nitrate at 45 °C. [1]

.... g per 100 g of water

II. Calculate the mass of solid potassium nitrate that forms when a saturated solution in 100 g of water at 45 °C cools to 25 °C. [1]

Mass = ... g

(c) Potassium nitrate decomposes on heating to produce potassium nitrite, KNO_2, and oxygen.

(i) Balance the equation for the reaction. [1]

$$2KNO_3 \longrightarrow \boxed{}\,KNO_2 + O_2$$

(ii) On heating potassium nitrate, a student expected to make 4.35 g of potassium nitrite. However, in an experiment she only made 3.13 g.

Calculate the percentage yield of potassium nitrite in her experiment. Give your answer to the nearest **whole** number. [2]

Percentage yield = ... %

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